

Sovereign Bond Markets, 2016–2026

Yields, volatility, debt stocks, and policy-rate cycles across major economies.

SEPTEMBER 2026 · QUANTITATIVE REPORT

*Computed from primary data via the ElkasabgiData Econ Data Library, 23 September 2026.

All yield, volatility, correlation, policy-cycle, and debt figures below are computed from the series themselves, not quoted from press. Issuance, QE/QT, ratings, and foreign-ownership figures are web research (OECD, press), clearly labeled as such.*

0. Methodology

Input	Source	Series	Frequency	Vintage
10Y sovereign bond yields	IMF International Financial Statistics (IFS)	FIGB_PA: “Interest Rates, Government Securities, Government Bonds”	Monthly	Ends 2024-12 to 2025-06 by country
10Y yields, US	Federal Reserve Board H.15	RIFLFCY10_N.M (10Y Treasury constant maturity)	Monthly	Through 2026-08
10Y yields, Germany	Deutsche Bundesbank	BBSIS series (inländische Inhaberschuldverschreibung en, RLZ 9–10y)	Monthly	Through 2026-08
10Y yields, FR/SE/NO/DK/FI/NL/JP	Sveriges Riksbank (reference series for foreign govt bonds)	RIKSBANK:*GVB10Y	Daily → monthly	Through 2026-09-21/22
10Y yields, Australia	Reserve Bank of Australia	FCMYGBAG10 (capital-market yields, 10Y govt bond)	Monthly	Through 2026-08
10Y yields, Canada	Bank of Canada	BD.CDN.10YR.DQ.YLD (benchmark bond yield)	Daily → monthly	Through 2026-09-21
Euro-area AAA 10Y spot curve	ECB Statistical Data Warehouse	YC:B.U2.EUR.4F.G_N_A.SV_C_YM.SR_10Y	Daily → monthly	Through 2026-09-21
Central-bank policy rates	BIS (WS_CBPOL)	End-of-month policy rates	Monthly	Through 2026-04-01 (mostly)
Government debt stocks	Eurostat sdg_17_40 (general govt, EU15); World Bank WDI GC.DOD.TOTL.GD.ZS (central govt, 19); IMF Historical Public Debt (general govt, 16)	Annual	Ends 2015 (HPD) to 2025 (Eurostat)	
Gross debt issuance	Not available as a time series in the library; OECD Global Debt Report 2025/2026 headline numbers from the earlier web-research pass, clearly labeled			

Volatility = annualized standard deviation of monthly yield changes ($\times\sqrt{12}$); “12m rolling vol” uses a 12-month rolling window. Correlations are Pearson correlations of monthly yield changes (daily for the COVID window). Cross-country dispersion = cross-sectional standard deviation of yield levels each month.

Coverage caveat. The library is wired database-by-database. Gaps remain: no monthly 10Y series in the library for China, India, Brazil, Mexico, Turkey, Russia, Argentina, Indonesia, Saudi Arabia, UAE, Taiwan,

Israel, Singapore, Thailand, Malaysia, Philippines, Vietnam, Pakistan, Iran, Egypt, Colombia, Peru, Chile, South Africa, Poland, Czechia, Romania, Portugal, Spain, Austria, Belgium, Ireland, Hong Kong, New Zealand, or Kazakhstan (see §1a). Figures are computed only over available data; nothing is imputed or invented.

1. Data coverage

1a. 10Y yields: available monthly series

Economy	Source	Coverage	End
United States	Fed Board H.15	1953-04 → 2026-08	2026-08
Germany	Bundesbank	1973-04 → 2026-08	2026-08
France	Riksbank ref.	1988-02 → 2026-09	2026-09-21
Japan	Riksbank ref.	1987-01 → 2026-09	2026-09-22
Sweden	Riksbank	1987-01 → 2026-09	2026-09-21
Norway	Riksbank ref.	1990-05 → 2026-09	2026-09-21
Denmark	Riksbank ref.	1982-01 → 2026-09	2026-09-21
Finland	Riksbank ref.	1990-04 → 2026-09	2026-09-21
Netherlands	Riksbank ref.	1987-02 → 2026-09	2026-09-21
Australia	RBA	2013-05 → 2026-08	2026-08
Canada	Bank of Canada	2001-01 → 2026-09	2026-09-21
United Kingdom	IMF IFS FIGB_PA	1957-01 → 2025-06	2025-06
Italy	IMF IFS FIGB_PA	1980-01 → 2025-06	2025-06
Morocco	IMF IFS FIGB_PA	1997-12 → 2025-06	2025-06
South Korea	IMF IFS FIGB_PA	1973-05 → 2025-05	2025-05
Switzerland	IMF IFS FIGB_PA	1998-01 → 2024-12	2024-12
Bangladesh	IMF IFS FIGB_PA	2006-08 → 2025-06	2025-06
Iceland	IMF IFS FIGB_PA	1992-01 → 2025-06	2025-06
Euro area (AAA curve)	ECB	2004-09 → 2026-09	2026-09-21
Stale IFS (Australia →2019-11, Canada →2017-04, India →2017-05, Japan →2017-05)	superseded by national sources where available		

Gaps (no monthly 10Y series found in the library): China, India, Brazil, Mexico, Turkey, Russia, Argentina, Indonesia, Saudi Arabia, UAE, Taiwan, Israel, Singapore, Thailand, Malaysia, Philippines, Vietnam, Pakistan, Iran, Egypt, Colombia, Peru, Chile, South Africa, Poland, Czechia, Romania, Portugal, Spain, Austria, Belgium, Ireland, Hong Kong SAR, New Zealand, Kazakhstan.

1b. Policy rates: BIS WS_CBPOL (monthly, mostly through 2026-04-01)

33 series: AR AU BR CA CH CL CN CO CZ DK EA GB HK ID IL IN JP KR MX MY NO PE PH PL RO RU SA SE TH TR US ZA. Vintage warning: the series end 2026-04-01, so they do not capture the Fed's 16 Sep 2026 hike to 3.75–4.00% (web-research).

1c. Debt stocks: 50/50 countries covered (3 sources)

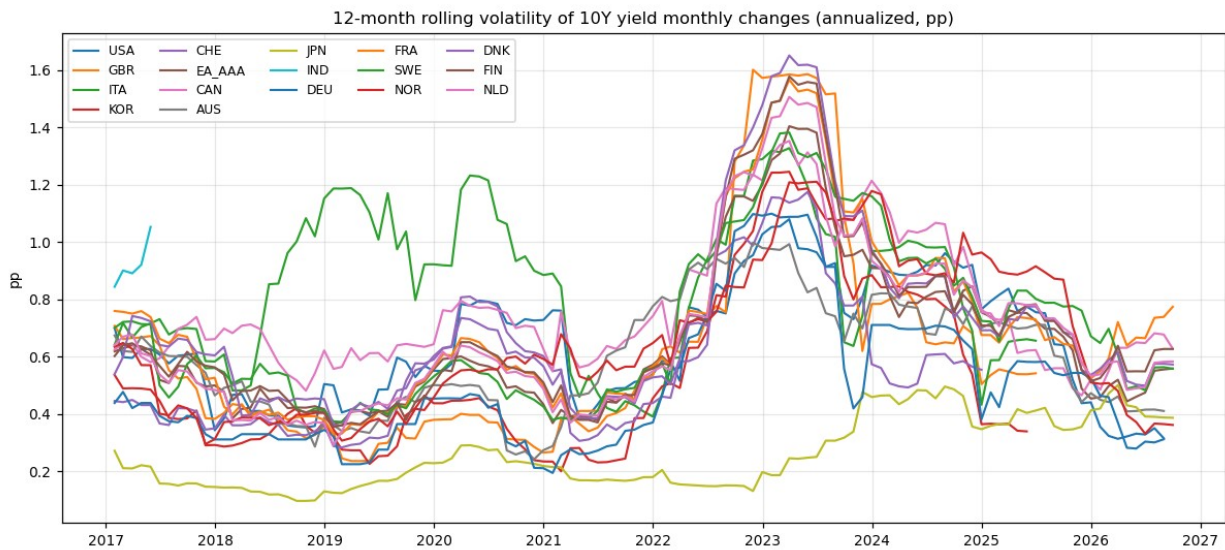
Source	Countries	Definition	Vintage
Eurostat sdg_17_40	15 EU economies (DE FR IT ES NL PL BE IE AT SE DK FI PT CZ RO)	General govt gross debt, % GDP (Maastricht)	2000–2025
World Bank WDI GC.DOD.TOTL.GD.ZS	19 (US GB CA AU IN BR RU KR MX SG CH TR TH MY CO ZA KZ PE PH)	Central govt debt, % GDP	Mostly 2024 (AU 2022, IN 2018, PE 2021, PH 2014, KZ 2023)

Source	Countries	Definition	Vintage
IMF Historical Public Debt HPD	16 (CN JP SA TW HK IR AR AE VN BD IL ID NO PK EG CL)	General govt gross debt	Ends 2015; no 2016+ data in the library

Sector mismatch matters: WDI is central-government only; Eurostat and HPD are true general government. The WDI UK series is 155.6% in 2016, 194.7% in 2020, and 130.7% in 2024. These are the published WDI central-government values and differ from IMF and OECD general-government measures; UK and Singapore comparisons require attention to the source definition. No usable country-level series exists in IMF Fiscal Monitor (aggregate-only), BIS, or OECD within this library.

2. Findings

2.1 Volatility: the 2022 regime break is measurable everywhere

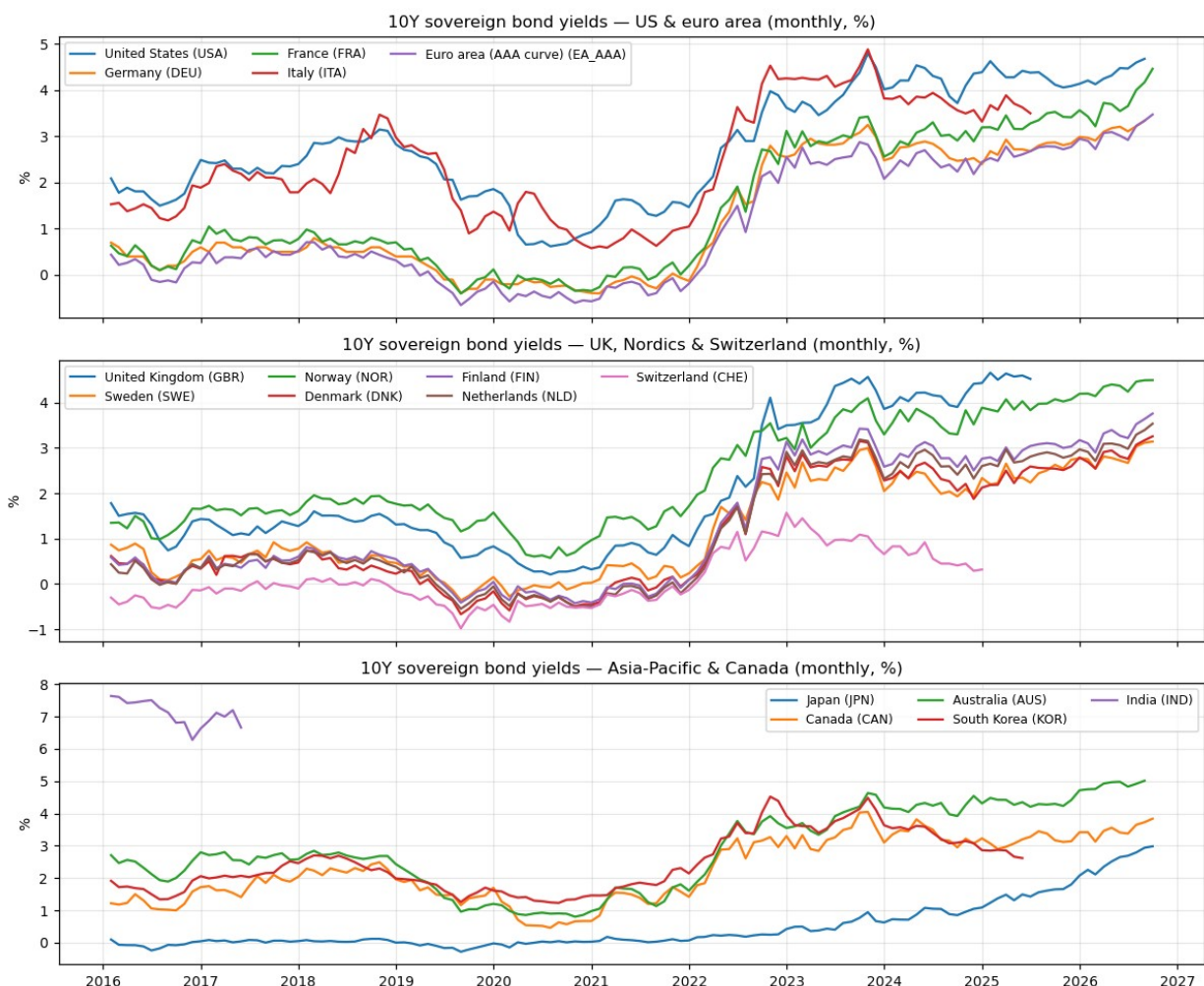


Annualized volatility of monthly 10Y-yield changes, 2022–2023 vs 2016–2019 (computed):

Economy	2016–2019	2022–2023	Multiple
Morocco	0.55	2.59	×4.7
United Kingdom	0.47	1.28	×2.7
Finland	0.50	1.31	×2.6
Germany	0.38	1.00	×2.6
South Korea	0.42	1.08	×2.5
Denmark	0.54	1.34	×2.5
Netherlands	0.51	1.26	×2.5
Iceland	0.42	1.04	×2.5
France	0.54	1.30	×2.4
Norway	0.46	1.08	×2.3
Euro area (AAA)	0.51	1.17	×2.3
Sweden	0.54	1.25	×2.3
Switzerland	0.44	0.94	×2.1
Canada	0.63	1.24	×2.0

Each market displayed in the volatility figure recorded 2.0 to 2.7 times its 2016–2019 yield volatility in 2022–2023. Italy was the exception in the full sample at 1.4 times: its 2016–2019 baseline was 0.91 and its 2022–2023 value was 1.28. Germany had the lowest baseline at 0.38 and a 2022–2023 value of 1.00.

2.2 Yield levels: the round trip (computed, 2016+)

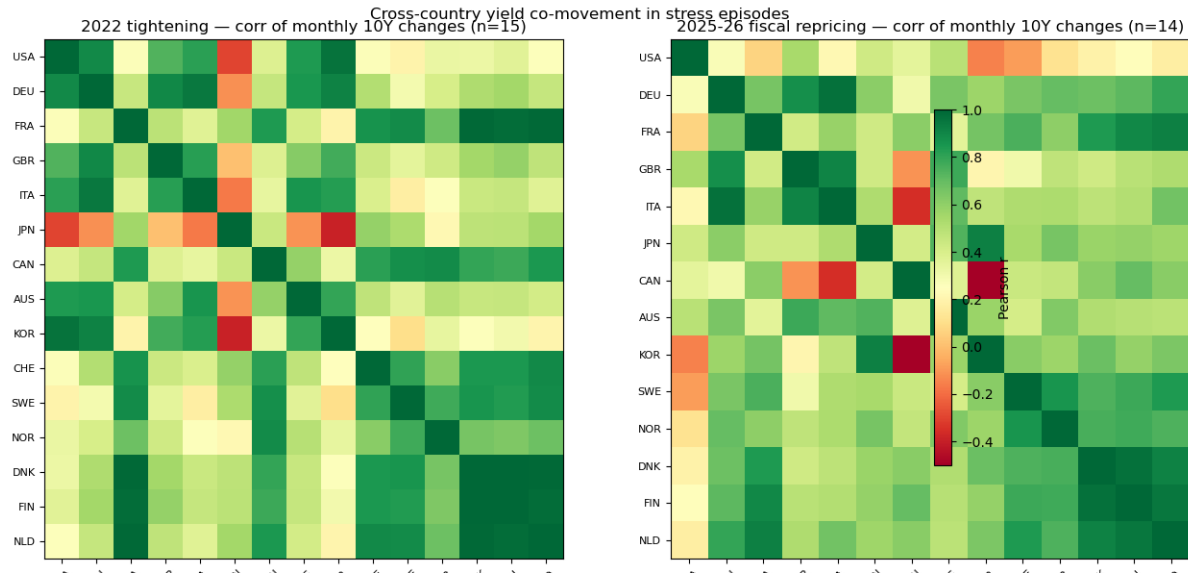


Major-market 10-year sovereign yields.

Economy	10Y min (date)	10Y max (date)	Net change 2016→latest
United States	0.62% (2020-07)	4.80% (2023-10)	+2.59pp (to 4.68%, 2026-08)
Germany	−0.40% (2019-08)	3.34% (2026-08)	+2.64pp
France	−0.40% (2019-08)	4.46% (2026-09)	+3.83pp
Japan	−0.28% (2019-08)	2.98% (2026-09)	+2.89pp
United Kingdom	0.21% (2020-07)	4.66% (2025-01)	+2.74pp (to 2025-06)
Italy	0.58% (2020-12)	4.89% (2023-10)	+1.97pp (to 2025-06)
Canada	0.46% (2020-07)	4.05% (2023-10)	+2.62pp
Australia	0.80% (2020-10)	5.02% (2026-08)	+2.31pp
South Korea	1.23% (2020-08)	4.52% (2022-10)	+0.70pp (to 2025-05)
Switzerland	−0.98% (2019-08)	1.57% (2022-12)	+0.62pp (to 2024-12)
Sweden	−0.36% (2019-08)	3.14% (2026-09)	+2.28pp
Norway	0.58% (2020-07)	4.50% (2026-09)	+3.15pp
Denmark	−0.67% (2019-08)	3.26% (2026-09)	+2.64pp
Finland	−0.43% (2020-10)	3.76% (2026-09)	+3.17pp
Netherlands	−0.55% (2019-08)	3.54% (2026-09)	+3.10pp
Euro area (AAA curve)	−0.65% (2019-08)	3.47% (2026-09)	+3.03pp

Across the observed markets, 10-year yields rose by approximately 3 percentage points from their 2019–2020 troughs. France’s 10-year yield reached 4.46% in September 2026, approximately 1.1 percentage points above Germany. Japan’s 10-year yield reached 2.98% in September 2026.

2.3 Stress episodes: co-movement was strongest in the COVID dash-for-cash



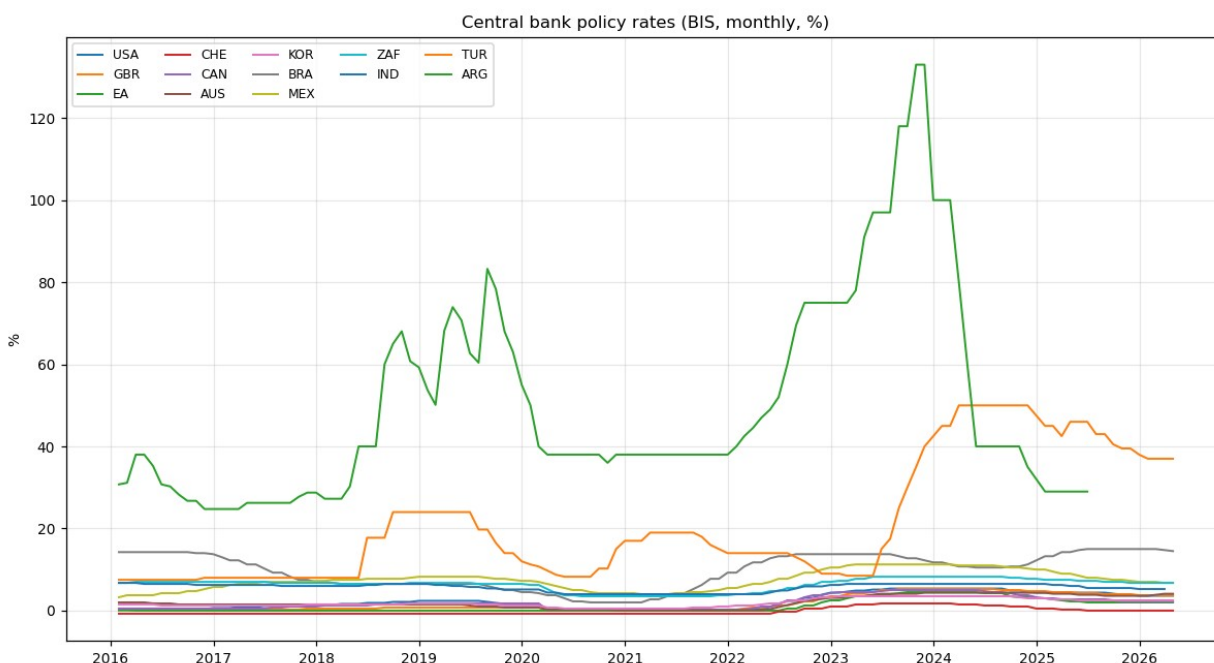
Mean pairwise correlation of yield changes across major markets (computed):

Episode	Countries	Mean r	Median r
COVID dash-for-cash (daily, 15 Feb–30 Apr 2020)	9	0.62	0.66
2022 tightening (monthly, Jan–Dec 2022)	15	0.54	0.51
2025–26 fiscal repricing (monthly, Jan 2025–Aug 2026)	14	0.54	0.57

The March 2020 cash-market disruption produced the highest co-movement. Euro sovereign yields rose together in March (Denmark +0.38pp, Finland +0.31pp, and France +0.29pp), while Canada (−0.41pp) and Norway (−0.21pp) declined as policy-rate expectations changed. Mean pairwise correlation was approximately 0.54 in both 2022 and 2025–2026. The latter episode was more concentrated among European markets and the United States.

Cross-country dispersion of major-market yield levels peaked at 1.22pp in October 2023 and declined to 0.73pp in 2026.

2.4 Policy-rate cycles: timing and subsequent reductions (BIS, computed)



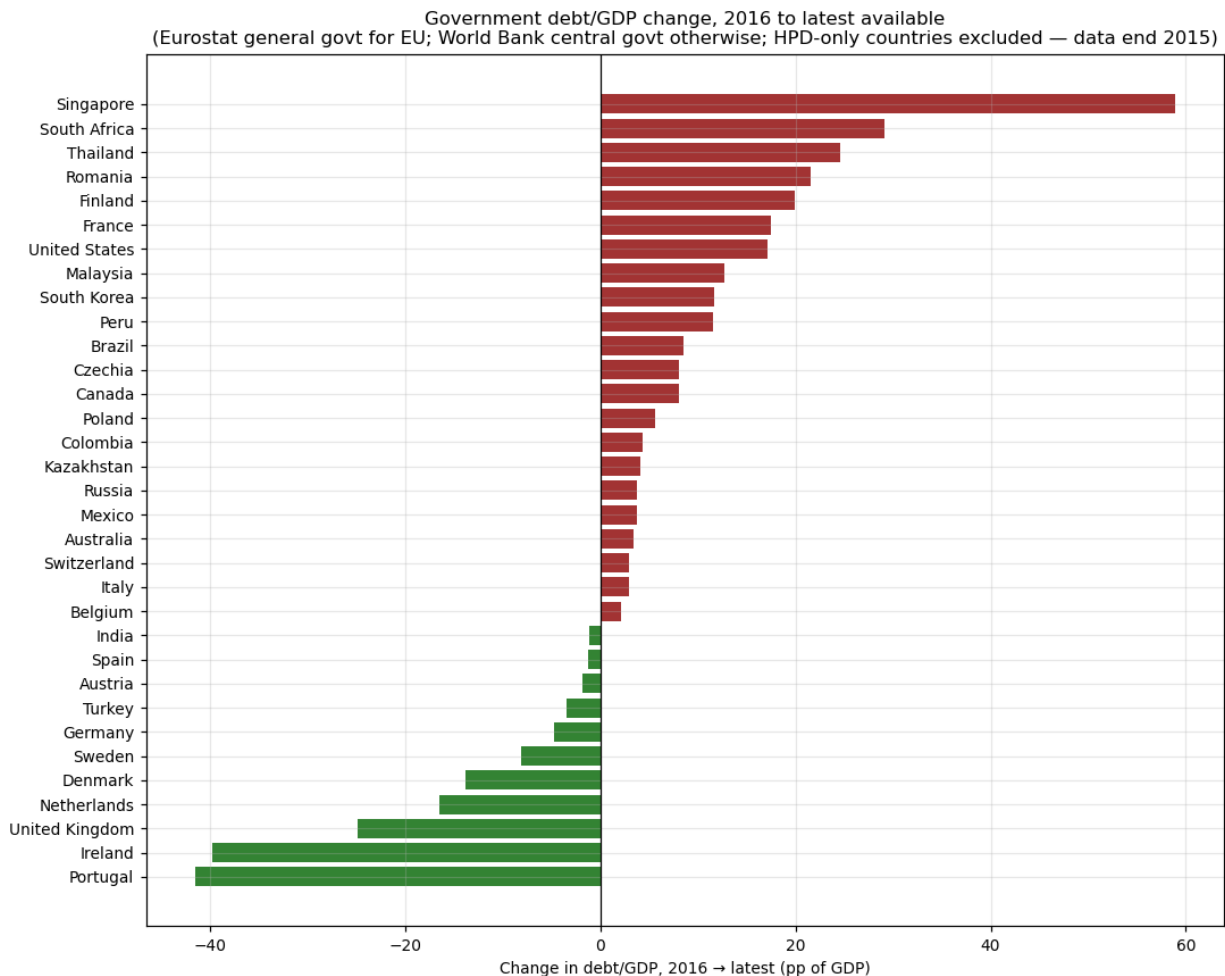
Policy-rate paths for the observed economies.

Trough → peak → latest (monthly):

Economy	Trough	Peak	Hike	Latest (2026-04)	Cut so far
United States	0.125% (2020-03)	5.375% (2023-07)	5.25pp	3.625%	1.75pp
Euro area	0.0% (2016-03)	4.5% (2023-09)	4.5pp	2.0%	2.5pp
United Kingdom	0.1% (2020-03)	5.25% (2023-08)	5.15pp	3.75%	1.5pp
Canada	0.25% (2020-03)	5.0% (2023-07)	4.75pp	2.25%	2.75pp
Australia	0.1% (2020-11)	4.35% (2023-11)	4.25pp	4.1%	0.25pp
Switzerland	−0.75% (2016-01)	1.75% (2023-06)	2.5pp	0.0%	1.75pp
Sweden	−0.5% (2016-02)	4.0% (2023-09)	4.5pp	1.75%	2.25pp
Norway	0.0% (2020-05)	4.5% (2023-12)	4.5pp	4.0%	0.5pp
Denmark	−0.75% (2019-09)	3.6% (2023-09)	4.35pp	1.6%	2.0pp
Japan	−0.1% (2016-09)	0.75% (2025-12)	0.85pp	0.75%	0.0pp
South Korea	0.5% (2020-05)	3.5% (2023-01)	3.0pp	2.5%	1.0pp
Brazil	2.0% (2020-08)	15.0% (2025-06)	13.0pp	14.5%	0.5pp
Mexico	3.25% (2016-01)	11.25% (2023-03)	8.0pp	6.75%	4.5pp
Chile	0.5% (2020-04)	11.25% (2022-10)	10.75pp	4.5%	6.75pp
Colombia	1.75% (2020-09)	13.25% (2023-05)	11.5pp	11.25%	2.0pp
Peru	0.25% (2020-04)	7.75% (2023-01)	7.5pp	4.25%	3.5pp
South Africa	3.5% (2020-07)	8.25% (2023-05)	4.75pp	6.75%	1.5pp
Turkey	7.5% (2016-01)	50.0% (2024-03)	42.5pp	37.0%	13.0pp
Argentina	24.75% (2017-01)	133.0% (2023-10)	108.25pp	29.0%	104.0pp
Russia	4.25% (2020-07)	21.0% (2024-10)	16.75pp	14.5%	6.5pp
China	4.35% (2016-01)	not applicable	not applicable	3.0%	1.35pp from 2016

Policy-rate peaks were concentrated between July 2022 and December 2023, but subsequent reductions differed. Japan remained at its 0.75% peak; Norway had cut 0.5pp and Brazil 0.5pp; Chile had cut 6.75pp and Mexico 4.5pp. Brazil reached its 15.0% peak in June 2025. BIS data end on 1 April 2026 and therefore exclude the Federal Reserve's 16 September 2026 increase to 3.75–4.00%.

2.5 Debt stocks: changes since 2016 (computed)



Change in government debt as a share of GDP.

Debt/GDP changes, 2016 → latest available:

Biggest rises: Singapore +58.9pp (108.9 → 167.8, central govt), South Africa +29.1pp (53.7 → 82.8), Thailand +24.6pp (37.3 → 61.9), Romania +21.5pp (37.8 → 59.3), Finland +19.9pp (68.6 → 88.5), France +17.5pp (98.1 → 115.6), US +17.1pp (98.7 → 115.8, to 2024), Malaysia +12.7pp, South Korea +11.7pp, Peru +11.5pp.

Biggest falls: Portugal -41.5pp (131.2 → 89.7), Ireland -39.8pp (72.7 → 32.9), Netherlands -16.5pp (60.9 → 44.4), Denmark -13.8pp (41.7 → 27.9), Sweden -8.1pp, Germany -4.8pp (68.3 → 63.5), Spain -1.3pp (102.0 → 100.7), Italy +2.9pp (134.2 → 137.1).

No 2016+ data exists in the library for China, Japan, Saudi Arabia, Taiwan, Hong Kong, Iran, Argentina, UAE, Vietnam, Bangladesh, Israel, Indonesia, Norway, Pakistan, Egypt, or Chile (IMF HPD series end in 2015); their 2015 values: Japan 248.0%, Egypt 89.0%, Israel 64.1%, Pakistan 63.6%, Vietnam 58.3%, Argentina 52.1%, China 42.9%, Taiwan 36.5%, Bangladesh 31.7%, Indonesia 27.3%, Norway 27.9%, UAE 18.1%, Chile 17.5%, Iran 15.9%, Saudi Arabia 5.0%, Hong Kong 0.1%.

Debt ratios increased in the United States, France, South Africa, Thailand, and Romania, raising the stock exposed to refinancing at current yields. Ratios declined in Portugal, Ireland, the Netherlands, Denmark, and Germany. Japan recorded approximately 248% of GDP in the 2015-vintage series.

2.6 Debt issuance: refinancing share of gross borrowing

(OECD Global Debt Report 2025/2026; the library contains no issuance time series.)

- OECD gross sovereign borrowing: record \$17tn in 2025 (~\$18tn projected 2026), up from \$12tn in 2022.
- Refinancing $\approx$ \$13.5tn (~80%) of gross; one-third of OECD fixed-rate debt matures by 2027; 60% was issued pre-2022 at near-zero rates.
- The US is ~70% of OECD refinancing needs (~\$9.5tn).

2.7 Other factors (web-research, labeled)

- QE fully unwound: central-bank share of OECD sovereign debt 29% $\rightarrow$ 19%; Fed QT ended Dec 2025; ECB reinvestments stopped; BoE voted Sep 2026 to run its gilt book to zero.
- September 2026 multi-decade yields: Fed hiked to 3.75–4.00% on 16 Sep 2026; US 10Y >5% (first since Jul 2007); Bund 3.52%; gilt ~5.3% (30Y 5.96%, since 1998); JGB ~3.0% (since 1996). Not in the computed data except the ECB AAA curve (3.47% on 2026-09-21); the Fed Board monthly US series ends 2026-08 at 4.68%.
- Ratings migration: US lost its last AAA (Moody's, May 2025); France to single-A; Poland downgraded Sep 2026; Russia default 2022; South Africa's first upgrade in ~20 years.
- Foreign ownership decline: China's UST holdings \$1.32tn (2013) $\rightarrow$ ~\$618bn (Jul 2026); foreigners hold ~30% of US marketable debt vs ~50% in 2008.

3. DM vs EM divergence

Among the 15 major markets, 2022–2023 volatility multiples ranged from 2.0 to 2.7 in both developed markets and the included emerging markets. Policy rates and debt ratios differed more substantially. Developed-market central banks other than Japan and Norway had reduced rates by 1–3.5pp from their peaks; Brazil and Turkey remained at double-digit rates. Peak policy rates were 133% in Argentina and 50% in Turkey. Debt-to-GDP ratios rose by 29pp in South Africa and 25pp in Thailand, while they fell by 41pp in Portugal and 40pp in Ireland.

4. Sources and citations (producer-first)

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- OECD Global Debt Report 2025 / 2026 (issuance section, via web research).

5. Caveats and data quality

1. Vintages differ. IFS yields end 2024-12 to 2025-06; national yields run to 2026-08/09; BIS policy rates end 2026-04-01 (miss the Sep 2026 Fed hike); Eurostat debt runs to 2025; WDI debt to 2024 (some countries earlier); IMF HPD debt ends 2015.
2. Sector definitions differ between debt sources (central vs general government); cross-country debt comparisons should be read within-source.
3. UK WDI debt values (155.6% in 2016) are as the World Bank publishes them for central government and differ from IMF/OECD general-government figures.
4. IFS FIGB_PA is the IMF's general "government bond yield" concept; national benchmark 10Y series take priority wherever available.
5. September 2026 market prints (US 10Y >5%, Fed hike, BoE gilt-runoff vote, Poland downgrade) come from the web-research pass, not from the API series.
6. Figures are computed over available data only; no imputation.